2) Write a python program to implement DES (Data Encryption Standard) Algorithm.

Algorithm: DES Encryption

1. **Start**
2. Read the plaintext message.
3. Read the secret key (64 bits / 8 bytes).
4. Convert plaintext into binary and divide it into 64-bit blocks.
5. Apply **Initial Permutation (IP)** on each block.
6. Divide the block into two halves:
   - Left half (L)
   - Right half (R)
7. Perform **16 rounds** of the following steps:
   1. Expand R from 32 bits to 48 bits.
   2. XOR the expanded R with the round key.
   3. Apply **S-Box substitution** (48 bits → 32 bits).
   4. Apply **Permutation (P-Box)**.
   5. XOR the result with L.
   6. Swap L and R.

Combine final L and R.
Apply **Final Permutation (FP)**.
Obtain the encrypted ciphertext.
Stop

Algorithm: DES Decryption

1. **Start**
2. Read the ciphertext.
3. Use the same secret key.
4. Apply **Initial Permutation (IP)**.
5. Perform **16 rounds in reverse order** using the round keys.
6. Combine the left and right halves.
7. Apply **Final Permutation (FP)**.
8. Obtain the original plaintext.
9. **Stop**

Program:

```
from Crypto.Cipher import DES
from Crypto.Util.Padding import pad, unpad
# Key must be exactly 8 bytes for DES
key = b'ABCDEFGH'
# Create DES object in ECB mode
des = DES.new(key, DES.MODE_ECB)
# Input message
plaintext = input("Enter Plain Text: ")
# Encryption
```

```
ciphertext = des.encrypt(pad(plaintext.encode(), 8))
print("Encrypted Text:", ciphertext)
# Decryption
decrypted_text = unpad(des.decrypt(ciphertext), 8)
print("Decrypted Text:", decrypted_text.decode())
```

Output:

```
Enter Plain Text: HELLO
Encrypted Text: b'\x8f\xa2...'
Decrypted Text: HELLO
```